

3.2 Annuities - Immediate

Annuities lasting n periods with payments at the end of each interest period is an

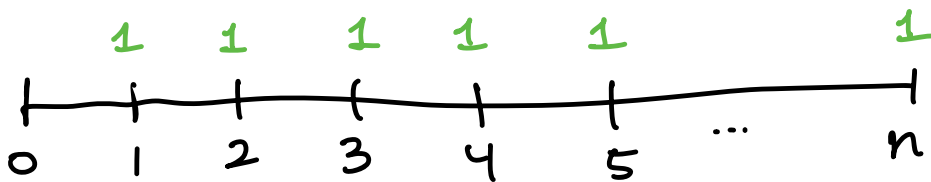
Annuity Immediate

why immediate?

Definition. A basic annuity Immediate has level payments of 1\$

The skeleton of every level payment annuity Immediate.

PV of basic annuity Immediate:



$$= 1v(1) + 1v(2) + \dots + 1v(n)$$

If $v(t) = (1+i)^{-t}$ (Compound Interest)
 $= v^t$

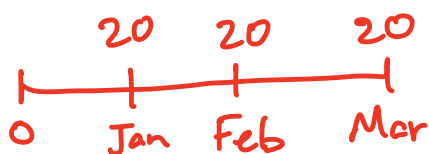
$$v + v^2 + v^3 + v^4 + \dots + v^n$$

$$= \frac{1}{1-v} - 1 - \left(v^{n+1} \left(\frac{1}{1-v} \right) \right)$$

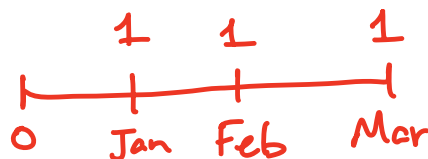
$$= \frac{1}{1-v} - \frac{1-v}{1-v} - \frac{v^{n+1}}{1-v} =$$

$$\boxed{\frac{v - v^{n+1}}{1-v}}$$

Problem: Marguerite receives an allowance of \$20 at the end of each month. Her four-year old brother Stevie receives an allowance of \$1 at month's end. Find the value on January 1 of each of their allowances for the three-month period consisting of January, February, and March, using a monthly effective interest rate of .5%.



M



S

$$\frac{i^{(12)}}{12} = .005$$

$$M: 20(1.005^{-1} + 1.005^{-2} + 1.005^{-3})$$

$$S: 1(\quad \quad \quad)$$

$$M = 59.405$$

$$S = 2.970248$$

Lets go back to the formula

$$PV(\text{basic annuity Immediate}) = \frac{v(1-v^n)}{1-v} = \boxed{\frac{1-v^n}{i}}$$

this is so common it gets its own symbol

$$a_{\overline{n}|}$$

when compound interest is used and the effective interest rate per payment is i we write

$$\boxed{a_{\overline{n}|i} = \frac{1-v^n}{i}}$$

Gives the value of the annuity
one payment before the 1st payment.

It is also useful to know the value of a basic annuity immediate lasting n period after the final payment. we use

$$S_{\overline{n}|} \quad \text{or} \quad S_{\overline{n}|i}$$

"Accumulated Value"

obvious fact : $S_{\overline{n}|i} = a_{\overline{n}|i} (1+i)^n$

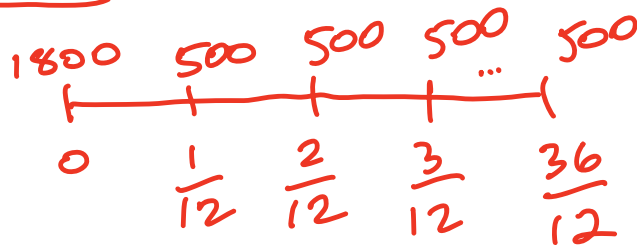
(More generally $S_{\overline{n}|} = a_{\overline{n}|} \cdot a(n)$)

Thus,

$$S_{\overline{n}|i} = \frac{(1+i)^n - 1}{i}$$

Problem: Candace wishes to buy a new car. Her budget allows her to make monthly payments of \$500, and she has saved \$1,800 for a down payment. Candace qualifies for a 36-month auto loan which has a nominal interest rate of 4.8% convertible monthly. How expensive a car can Candace buy?

Solution



$$i^{(12)} = .048$$

$$1800 + a_{\overline{36}|.004}$$

$$1800 + PV \left(500\$ \text{ annuity immediate} \right)$$

$$= 1800 + 500 \left(\frac{1 - \left(1 - \frac{d^{(12)}}{12} \right)^{36}}{\frac{.048}{12}} \right)$$

$$= 1800 + 500 \left(\frac{1 - \left(1 - \frac{.004}{1.004} \right)^{36}}{\frac{.048}{12}} \right)$$

$$= 18532.93795$$

Problem: Seth wishes to borrow \$4,400 so he can pay his tuition. He qualifies for a two-year loan with a monthly effective interest rate of .25% and level payments. Find the amount of Seth's monthly payments under this loan.

Solution.

$$\frac{.12}{12} = .0025$$

$$4400 = X \cdot a_{\overline{24}|.0025}$$

$$= X \left(\frac{1 - \left(1 - \frac{.0025}{.0025}\right)^{24}}{.0025} \right) \Rightarrow X = 189.12$$

because of rounding if some makes 24 pmts of \$189.12 they will over pay so the last pmt is often times adjusted.

calc PV of 23 pmts and missing amt at $t=24$.

Problem: Mr. Liang wishes to accumulate \$200,000 in a college fund for his newborn daughter. He wishes the money to be available to her on her eighteenth birthday, and he is prepared to make level contributions on her first through eighteenth birthdays. The fund has an annual effective discount rate of 5%. How large will Mr. Liang's contributions need to be?

Solution. $d = 5\% \Rightarrow i = \frac{.05}{.95}$
 $\Rightarrow v = .95$

$$200,000 = X S_{\overline{18}|i} = X a_{\overline{18}|i} (1+i)^{18}$$

$$= X (28.8331196)$$

$$X = 6936.467604$$

Problem: Hyun wishes to contribute \$100 at the end of each month to his savings account until the account has a balance of at least \$3,000, at which time he will buy his grandmother's piano. Hyun's savings account has an annual interest rate of 5.4% convertible monthly. How many months it will take Hyun to accumulate the needed money, and how much money will he then have saved?

solution. $i^{(12)} = .054 \Rightarrow \frac{i^{(12)}}{12} = .0045$

$$3000 = 100 S_{\overline{n}|.0045} \quad \frac{d^{(12)}}{12} = .004479841$$

$$30 = \left(\frac{1 - (1 - \frac{.0045}{1.0045})^n}{.0045} \right) (1.0045)^n$$

$$.135 = \left(1 - \left(1 - \frac{.0045}{1.0045}\right)^n\right)(1.0045)^n$$

$$.135 = 1.0045^n - \left((1.0045) \left(1 - \frac{.0045}{1.0045}\right)\right)^n$$

$$.135 = 1.0045^n - 1$$

$$n = \ln(1.135) / \ln(1.0045) = 28.2$$

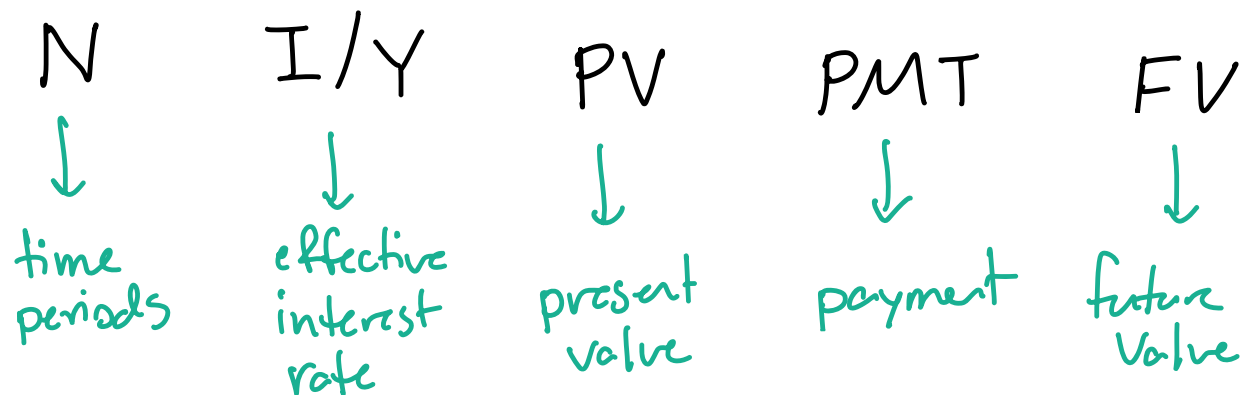
So 29 months

The following formula can be rederived
from $S_{\overline{n}|i} = a_{\overline{n}|i}(1+i)^n$

$$i + \frac{1}{S_{\overline{n}|i}} = \frac{1}{a_{\overline{n}|i}}$$

BAII - Plus can solve an annuity
immediate

When 4 of the 5 are known!



3.3 Annuities - Due

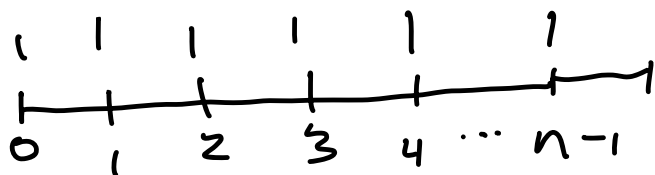
An annuity where payments are made at the beginning of each interest period is an annuity-due.

A basic annuity due has pmts of \$1. We use the following notation for its P.V. and F.V.

$$\ddot{a}_{\overline{n}|}$$

$$\ddot{s}_{\overline{n}|}$$

If $a(t) = (1+i)^t$ then



$$\begin{aligned}\ddot{a}_{\overline{n}|i} &= 1 + (1+i)^{-1} + (1+i)^{-2} + \dots + (1+i)^{-(n-1)} \\ &= 1 + v + v^2 + \dots + v^{n-1}\end{aligned}$$

$$= \sum_{k=0}^{\infty} (1+i)^{-k} - \sum_{k=n}^{\infty} (1+i)^{-k}$$

$$= \sum_{k=0}^{\infty} v^k - \sum_{k=n}^{\infty} v^k$$

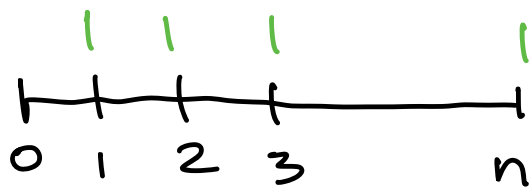
$$= \frac{1}{1-v} - \left(v^n \left(\frac{1}{1-v} \right) \right)$$

$$= \frac{1}{1-v} (1 - v^n) = \frac{1 - v^n}{1 - v}$$

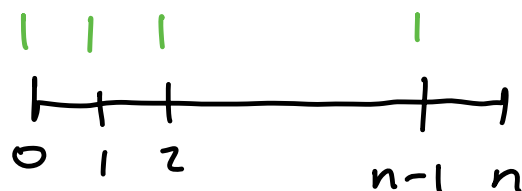
$$\ddot{a}_{\overline{n}|i} = \frac{1 - v^n}{1 - v} = \frac{1 - v^n}{d}$$

as always we have $\ddot{s}_{\overline{n}|i} = \ddot{a}_{\overline{n}|i} (1+i)^n$

Question: How are $a_{\overline{n}|i}$ and $\ddot{a}_{\overline{n}|i}$ related?



vs.



$$\ddot{a}_{\overline{n}|i} = a_{\overline{n}|i} + 1 - (1+i)^{-n}$$

$a_{\overline{n-1}|i}$

Two green arrows originate from the green text $a_{\overline{n-1}|i}$ below the equation. One arrow points to the $a_{\overline{n}|i}$ term in the equation, and the other points to the $-(1+i)^{-n}$ term.

$$\boxed{\ddot{a}_{\overline{n}|i} = a_{\overline{n-1}|i} + 1}$$

also

$$\ddot{a}_{\overline{n}|i} = 1 + (1+i)^{-1} + (1+i)^{-2} + \dots + (1+i)^{-n+1}$$

$$a_{\overline{n}|i} = (1+i)^{-1} + (1+i)^{-2} + \dots + (1+i)^{-n}$$

so $\ddot{a}_{\overline{n}|i} (1+i)^{-1} = a_{\overline{n}|i}$ or $\boxed{\ddot{a}_{\overline{n}|i} = (1+i) a_{\overline{n}|i}}$

Problem: During 1998, Omar had a dividend payment of \$100 directly deposited to his savings account on the first day of each month. Find the accumulated value of these payments at the end of the year if his savings account had a nominal discount rate of 4.8% payable monthly.

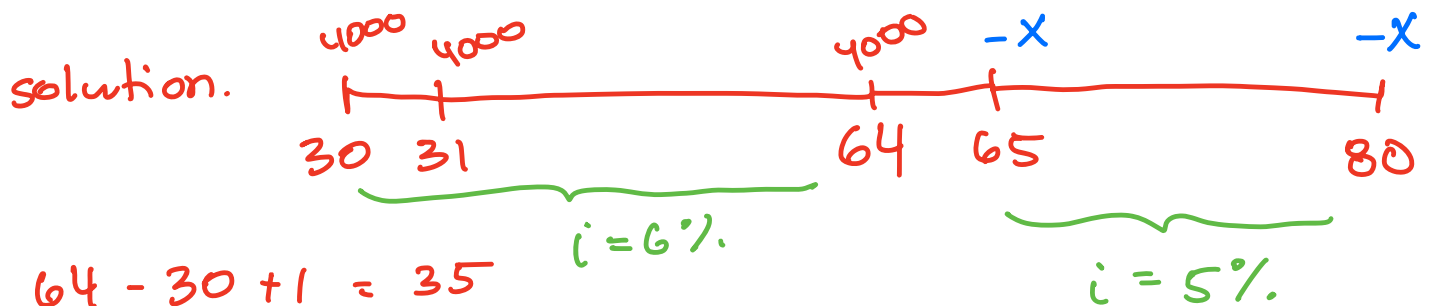
Solution. $pmt = 100$ $d^{(12)} = 4.8\%$

$$\frac{d^{(12)}}{12} = .004 \quad 100 \ddot{a}_{\overline{12}|.004} = 100 \left(\frac{1 - (.996)^{12}}{.004} \right)$$

$$= 1173.948852$$

So $\ddot{s}_{\overline{12}|.004} = 1231.79$

Problem: Dr. Hillary Street began making contributions to a new retirement account on her thirtieth birthday. She made a contribution of \$4,000 at the beginning of each year through her sixty-fourth birthday. Starting at age sixty-five and continuing through her eightieth birthday, she made a level withdrawal on her birthday. Find the amount of these withdrawals if they completely exhaust the balance in her account, and the annual effective interest rate is 6% until she is sixty-five, then 5% thereafter.



$64 - 30 + 1 = 35$

$$4000 \ddot{s}_{\overline{35}|.06} = X \ddot{a}_{\overline{16}|.05}$$

solving for X we get

$$X = \frac{4000 (1.06)^{35} \ddot{a}_{\overline{35}|0.06}}{\ddot{a}_{\overline{16}|0.05}} = 41,520.0057$$

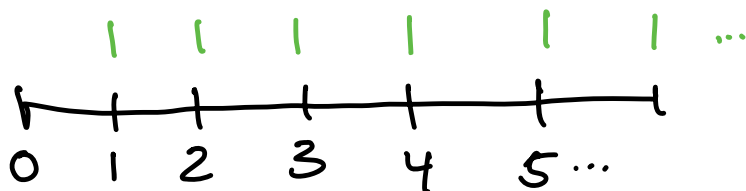
3.4 Perpetuities

A perpetuity is an annuity with an infinite term. We use

$a_{\infty|i}$ or $\ddot{a}_{\infty|i}$ to denote

the present value of a basic perpetuity
(of \$1)

obvious fact: $a_{\infty|i} = \ddot{a}_{\infty|i} - 1$



$$a_{\infty|i} = v + v^2 + v^3 + \dots = \frac{v}{1-v} = \frac{1}{i}$$

so

$$\ddot{a}_{\infty|i} = \frac{1}{i} + 1 = \frac{1+i}{i} = \frac{1}{d}$$

Problem: Gustave Larson has saved \$20,000. On January 1, he purchases a perpetuity with annual end-of-year payments. The perpetuity price is based on an annual effective interest rate of 5%. What are the annual payments for Gustave's perpetuity?

solution. $20000 = X \ddot{a}_{\infty|0.05} = \frac{X}{.05}$

$$\Rightarrow X = 1000$$

Problem: Norman wishes to leave an inheritance to three charities. The total inheritance is a series of level payments at the beginning of each year forever. He wishes charities A and B (which support medical research) to share the payments equally for ten years, and then all payments revert to charity C (that helps needy children). If the shares received by all three charities have the same value at the time of the bequest, find the annual effective interest rate i .

solution. perpetuity due.

$$\begin{aligned} A's \text{ charity} = B's \text{ charity} &= \frac{P}{2} \ddot{a}_{\overline{10}|i} \\ &= \frac{P}{2} \left(\frac{1 - v^{10}}{d} \right) \end{aligned}$$

$$C's \text{ charity} : P \cdot \ddot{a}_{\infty|i} = \frac{P}{d}$$

so starting at $t=10$

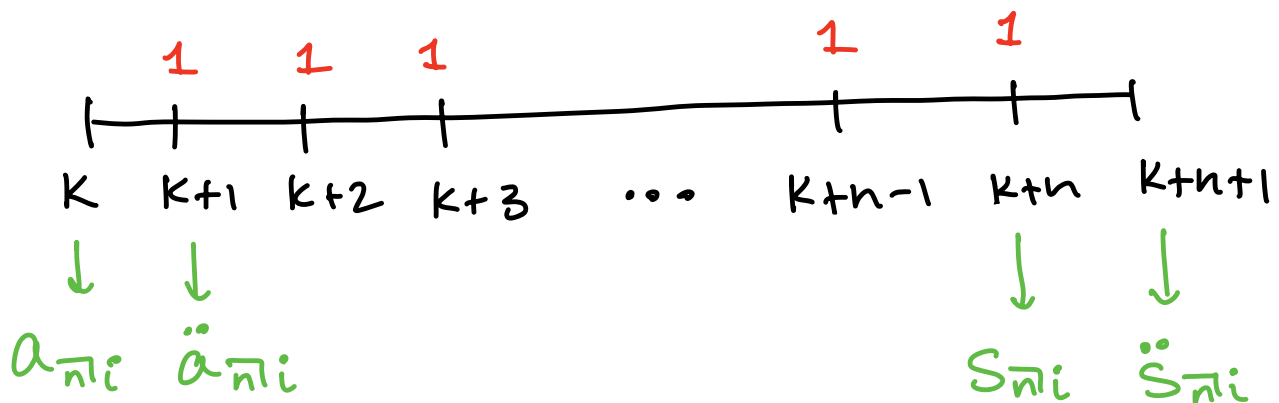
$$\frac{P}{d} v^{10} = \frac{P}{2} \left(\frac{1 - v^{10}}{d} \right) \Rightarrow 2v^{10} = 1 - v^{10}$$

$$\Rightarrow v^{10} = \frac{1}{3} \Rightarrow v = .89595846$$

$$\Rightarrow i = 11.6123174\%$$

3.5 Deferred Annuities

so far we've seen $a_{\overline{n}|i}$, $\ddot{a}_{\overline{n}|i}$, $s_{\overline{n}|i}$, and $\ddot{s}_{\overline{n}|i}$ which all give the value of \$1 n times just at different time periods

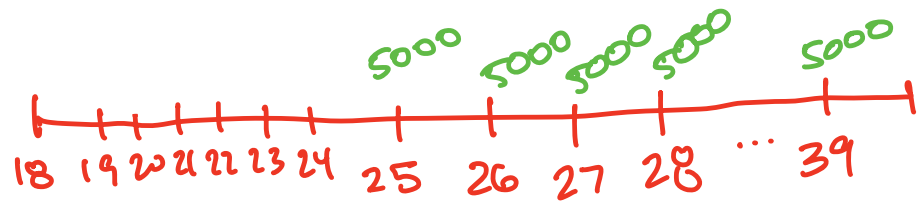


to get the value at other times we multiply by the appropriate power of $(1+i)$.

An alternative way to do this is to write these as a sum or difference of two (or more) annuities.

Problem: On her eighteenth birthday, Latisha receives an annuity that is to pay \$5,000 on her twenty-fifth through thirty-ninth birthdays. Calculate the value of the annuity on her eighteenth birthday using an annual effective interest rate of 5%.

Solution.



option 1: $5000 \ddot{a}_{\overline{17}|0.05} (1.05)^{-7}$

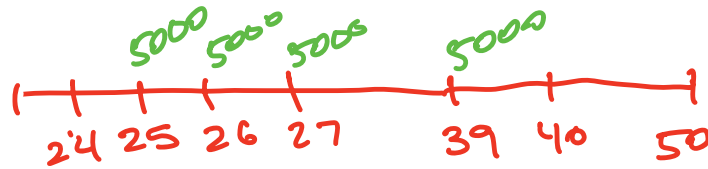
$$= 54,493.20470 (1.05)^{-7} = 38727.3032$$



$$5000 a_{\overline{27}|0.05} - 5000 a_{\overline{6}|0.05}$$

Problem: On her eighteenth birthday, Latisha receives an annuity that is to pay \$5,000 on her twenty-fifth through thirty-ninth birthdays. Calculate the value of the annuity on her fiftieth birthday using an annual effective interest rate of 5%.

Solution.



$$39 - 25 + 1 = 15$$

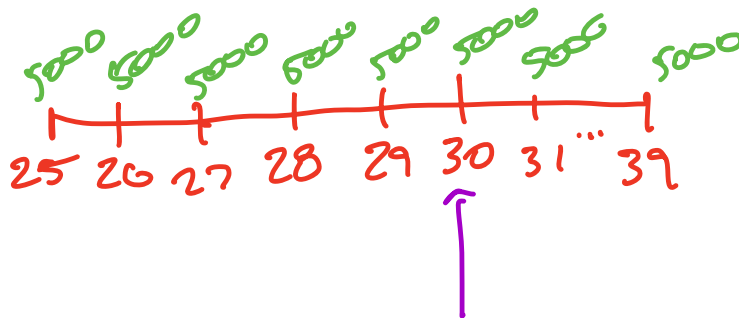
$$50 - 25 + 1 = 26$$

option 1: $a_{\overline{15}|5\%} (1.05)^{26} (5000) = 184,533.3329$

option 2: $(s_{\overline{26}|.05} - s_{\overline{11}|.05}) 5000.$

Problem: On her eighteenth birthday, Latisha receives an annuity that is to pay \$5,000 on her twenty-fifth through thirty-ninth birthdays. Calculate the value of the annuity on her thirtieth birthday using an annual effective interest rate of 5%.

Solution.



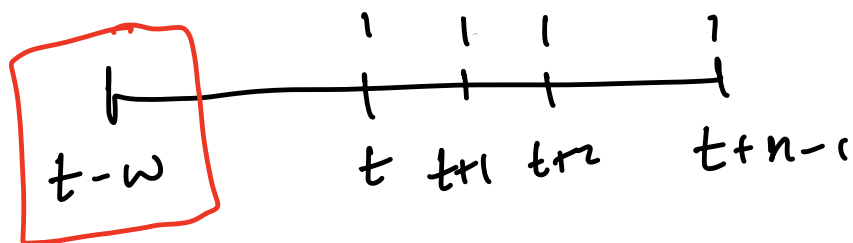
$$39 - 31 + 1 = 9$$

$$5000 + \ddot{s}_{\overline{31}|.05} (5000) + a_{\overline{9}|.05} (5000) = 69,548.67244$$

Sometimes the notation:

$w/n \ddot{a}$ is used to denote

value w periods before 1st pmt
for n pmts



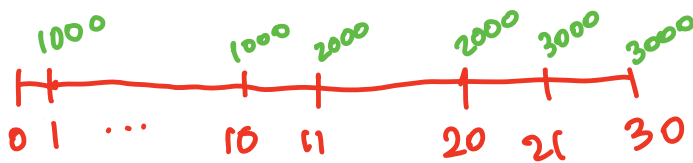
So $w/n \ddot{a}$ gives the value $w+1$
periods before 1st pmt.

This notation also works for perpetuities

$w/\infty \ddot{a}$ etc.

Problem: Marlene received a gift of a thirty-year annuity-immediate on the day she was born. The annuity pays \$1,000 on her first ten birthdays, \$2,000 on her next ten birthdays, and \$3,000 on the following ten birthdays. (i) If the nominal rate of interest is 5% convertible quarterly, find the value of the annuity on the day she was born. (ii) If the value of the annuity on the day she was born is \$25,000 when calculated using an annual effective rate of interest of j , find j correct to the nearest one-hundredth of a percent.

Solution.



$i^{(4)} = 5\%$
Find P.V.

(i)

$$1000 a_{\overline{10}|i} + 2000 a_{\overline{10}|i} (1+i)^{-10} + 3000 a_{\overline{10}|i} (1+i)^{-20} \\ = 25,575.19$$

(ii) $25000 = a_{\overline{10}|j} (1000 + 2000(1+j)^{-10} + 3000(1+j)^{-20})$

$$25000 = \left(\frac{1 - (1+j)^{-10}}{j} \right) (1000 + 2000(1+j)^{-10} + 3000(1+j)^{-20})$$

$$25000 j = 1000 + 2000(1+j)^{-10} + 3000(1+j)^{-20} - 1000(1+j)^{-10} \\ - 2000(1+j)^{-20} - 3000(1+j)^{-30}$$

$$\text{Let } (1+j)^{-10} = x \Rightarrow \frac{1}{(1+j)^{10}} = x \Rightarrow j = \left(\frac{1}{x} \right)^{1/10} - 1$$

$$25000j = 1000 + 1000x + 1000x^2 - 3000x^3$$

$$\Rightarrow j = .0525 \text{ using calculator!}$$

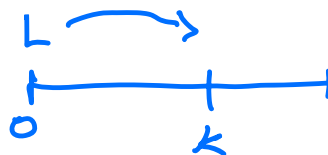
3.6 Outstanding Loans

Suppose a loan L is to be repaid at the end of n time periods by end of period pmts (Assume level) pmts

How do we find the loan balance some point between the loans origin and final pmt?

Method 1: Retrospective ←

If no pmts had occurred

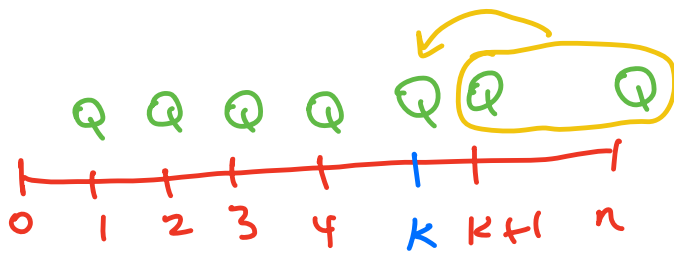


$$\text{Balance at } k = L(1+i)^k$$

but there could be pmts say Q so

$$\text{Balance}_k = L(1+i)^k - Qs_{\overline{k}|i}$$

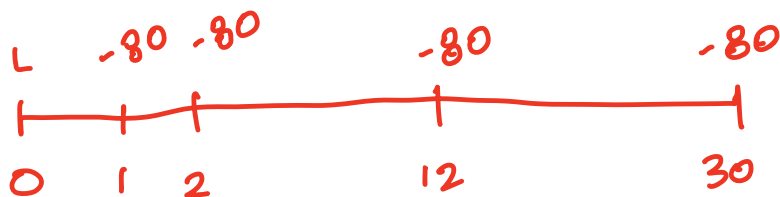
↑
time k



prospective method: $OLB_k = Qa_{\overline{n-k}|i}$

Problem: Sasha is obligated to repay a loan made on the first of the month by paying \$80 at the end of each of the next thirty months, including this month. The effective monthly rate on the loan is .4%. Find Sasha's loan balance immediately after her twelfth payment.

Solution.



$$\frac{i^{(12)}}{12} = .004$$

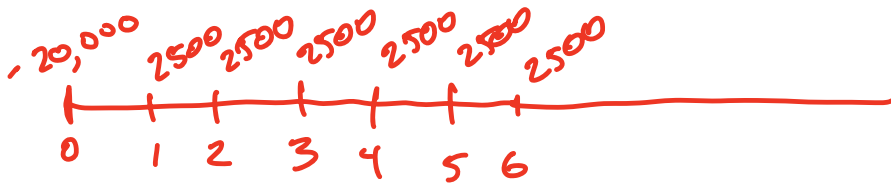
retrospective, $L - 80s_{\overline{12}|.004}$ don't know L...

prospective: $30 - 12 = 18$ pmts left

$$80 a_{\overline{18}|.004} = 80 \left(\frac{1 - \left(1 - \frac{.004}{1.004}\right)^{18}}{.004} \right) = 1386.71$$

Problem: A loan of \$20,000 is being repaid by payments of \$2,500 at the end of each year and a final smaller payment one year after the last \$2,500 payment. The annual effective interest rate on the loan is 8%. Find the outstanding loan balance just after the borrower has made payments totaling \$15,000.

Solution. $L = 20,000$, $pmts = 2,500$, $i = 8\%$



$$B_6 = 20,000 (1.08)^6 - 2500 s_{\overline{6}|.08}$$

$$= 20000(1.08)^6 - 2500(1.08)^6 \left(\frac{1 - (1 - \frac{.08}{1.08})^6}{.08} \right) = 13397.66$$

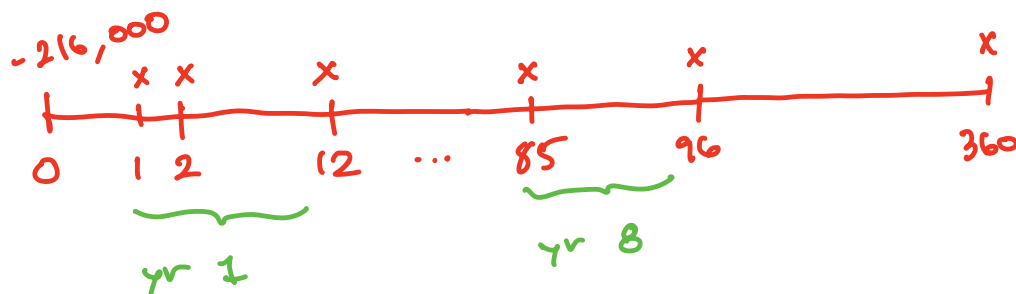
Problem: Mr. and Mrs. Harper purchase a home for \$256,000. They make a down payment of \$40,000 and finance the remainder of the purchase price with a thirty-year mortgage at an annual effective interest rate of 6.5%. The Harpers sell their home at the end of eight years, just after having made their ninety-sixth end-of-month mortgage payment. The sales price is \$282,000, and the Harpers closing costs are 3% of the selling price. The outstanding loan balance is deducted from the amount the Harpers receive and sent to the lender. How large a check do the Harpers receive, and how much interest did they pay over the eight years?

Solution. $L = 216,000$ down Pmt = 40,000

remainder = 216,000 30 yrs, $i = 6.5\%$.

monthly pmts

$$i^{(12)}_{/12} = (1.065)^{1/12} - 1$$



$$216000 = a_{\overline{360}|i^{(12)}_{/12}}(x) \Rightarrow x = 1338.95771314$$

$$B_8 = 216,000(1.065)^8 - 1338.95771314(1.065)^8 \left(\frac{1 - (1 - d^{(12)}_{/12})^{96}}{i^{(12)}_{/12}} \right)$$

$$= 190,800.52763$$

Interest:

$$.17(282,000) - B_8 = \underline{\underline{82739.76}}$$

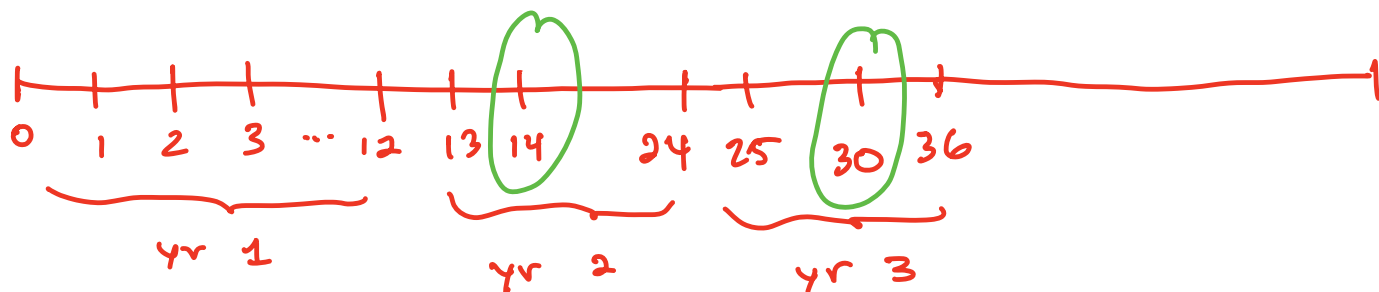
$$1338.9577(96) - (216,000 - 190,800.52763)$$

$$\underline{\underline{\approx 103,340.40}}$$

What happens if someone misses pmts?

Problem: Waswate took out a sixty-month car loan with an interest rate of 3% convertible monthly. To repay the loan, Waswate was to make monthly payments of \$252.65. During the first three years of the loan, he made all of the payments as scheduled except that he neglected to pay the fourteenth and thirtieth. What is the outstanding balance at the end of three years, just after he has made the payment scheduled at that time?

Solution. 60 mnth loan, $i^{(12)} = 3\%$, $pmt = 252.65$

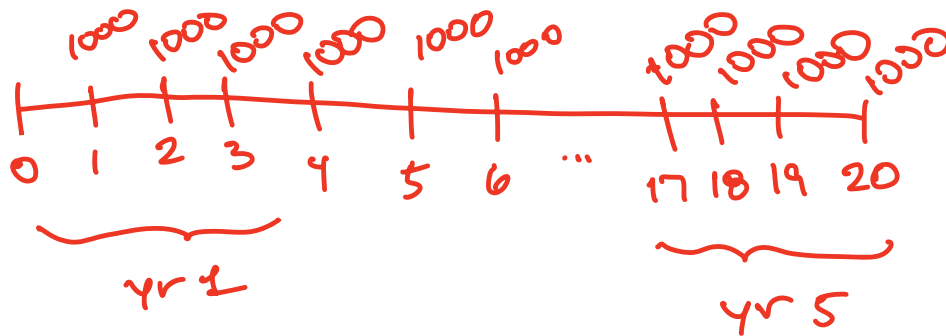


$$\begin{aligned}
 B_{36} &= 252 a_{\overline{60-36}|.03/12} + 252 \left(1 + \frac{.03}{12}\right)^{22} + 252 \left(1 + \frac{.03}{12}\right)^6 \\
 &= 252.65 \left(\frac{1 - \left(1 - \left(\frac{.03/12}{1 + .03/12}\right)^{24}\right)}{\frac{.03}{12}} + \left(1 + \frac{.03}{12}\right)^{22} + \left(1 + \frac{.03}{12}\right)^6 \right) \\
 &= 6401.53
 \end{aligned}$$

How about if rates vary?

Problem: Suppose that a loan is to be repaid by twenty end-of-quarter payments of \$1,000. The interest rate for the first two years is 6% convertible quarterly, and for the last three years it is 8% convertible quarterly. Find the loan amount, the outstanding loan balance just after the sixth payment, and the outstanding loan balance immediately after the fifteenth payment.

Solution. 20 end of quarter pmts (5 years)



yrs 1 and 2

$$i^{(4)} = 6\%$$

yrs 3, 4, 5

$$i^{(4)} = 8\%$$

9, 10, ..., 20

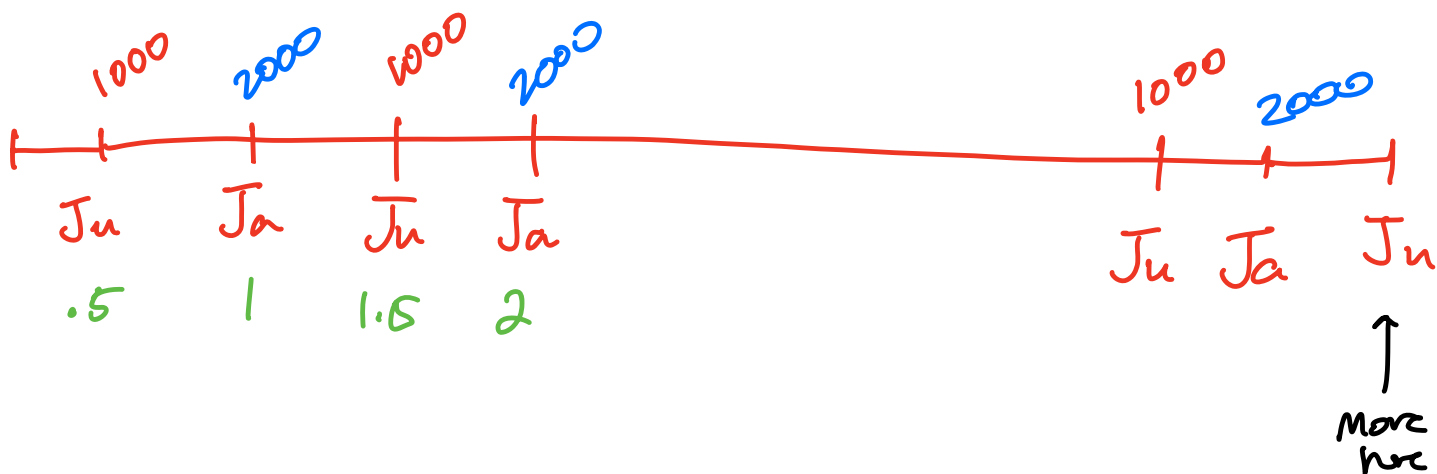
$$\begin{aligned}
 B_6 &= 1000 \cdot 3/a_{\overline{12}|.08/4} + 1000(1 + \frac{.06}{4})^{-1} + 1000(1 + \frac{.06}{4})^{-2} \\
 &= 1000 \left((1 + \frac{.06}{4})^{-2} \left(\frac{1 - (1 - \frac{.08/4}{1 + .08/4})^{12}}{.08/4} \right) + (1 + \frac{.06}{4})^{-1} + (1 + \frac{.06}{4})^{-2} \right) \\
 &= 12,220.96
 \end{aligned}$$

$$B_{15} = 1000 a_{\overline{57.08}|4} = 4713.46$$

3.7 Non-Level Annuities

How do we compute present and future values of annuities whose payments vary?

Imagine we won the lottery and got 20 years of payments of \$1000 on July 1 and \$2000 on January 1st. If we won on July 1st what is accumulated value after the 20 pmts? $i=5\%$

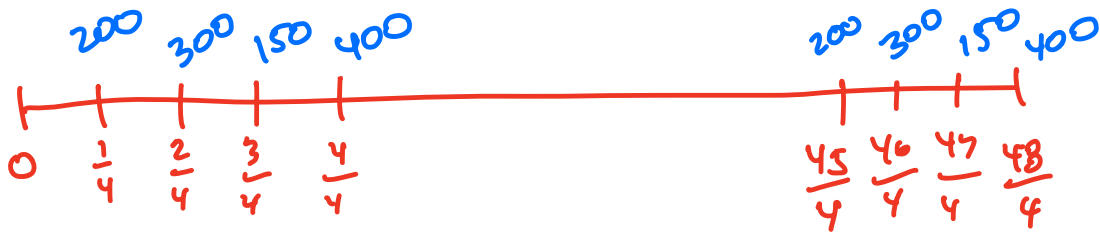


$$= 1000 \ddot{S}_{\overline{20}|.05} + 2000 S_{\overline{20}|.05} (1.05)^{1/2}$$

$$= 102,484.292552$$

Problem: An annuity lasts for twelve calendar years. At the end of each quarter, there is a payment. First quarter payments are \$200, second quarter payments are \$300, third quarter payments are \$150, and fourth quarter payments are \$400. Find the accumulated value of this annuity just after the last payment using a nominal quarterly interest rate of 7.2%.

Solution



$$i^{(4)} = 7.2\% \Rightarrow i = \left(1 + \frac{.072}{4}\right)^4 - 1$$

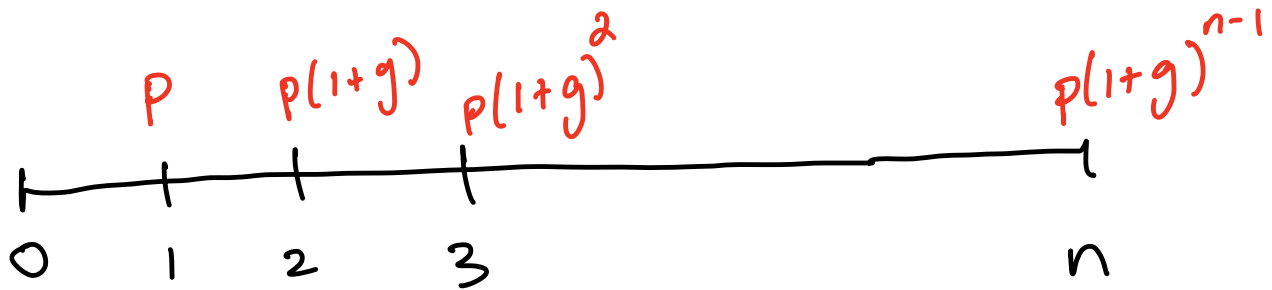
$$200 s_{\overline{12}|i} \left(1 + \frac{.072}{4}\right)^3 + 300 s_{\overline{12}|i} \left(1 + \frac{.072}{4}\right)^2$$

$$+ 150 s_{\overline{12}|i} \left(1 + \frac{.072}{4}\right)^1 + 400 s_{\overline{12}|i}$$

$$= s_{\overline{12}|i} \left(200 \left(1 + \frac{.072}{4}\right)^3 + 300 \left(1 + \frac{.072}{4}\right)^2 + 150 \left(1 + \frac{.072}{4}\right) + 400 \right)$$

$$= 19,677.64$$

3.8 Annuities with Geometric Progression Payments



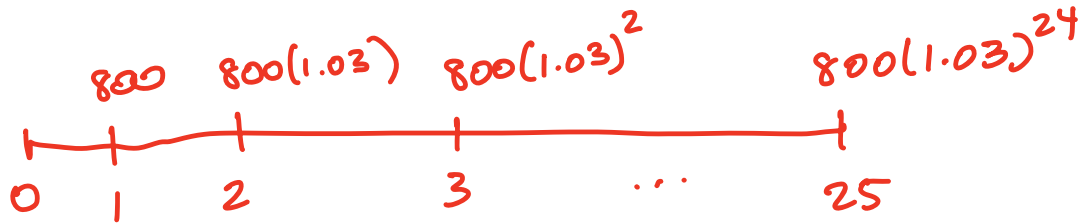
In this case, the present value would be :

$$\begin{aligned} & Pv + P(1+g)v^2 + P(1+g)^2v^3 + \dots + P(1+g)^{n-1}v^n \\ &= P \left[v + (1+g)v^2 + ((1+g)v)^2 + \dots + ((1+g)v)^{n-1} \right] \\ &= P \left[\frac{v(1 - ((1+g)v)^n)}{1 - (1+g)v} \right] = P \left[\frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i - g} \right] \end{aligned}$$

If $i = g \Rightarrow$ value is $nP(1+i)^{-1}$

Problem: On June 15, 1975, Roy purchased an annuity-immediate with annual payments for twenty-five years. The first payment was \$800 and the payments increased by 3% each year. The purchase price was based on an annual effective interest rate of 7%. Find this price.

Solution. $n = 25$, $P = 800$, $g = .03$, $i = .07$



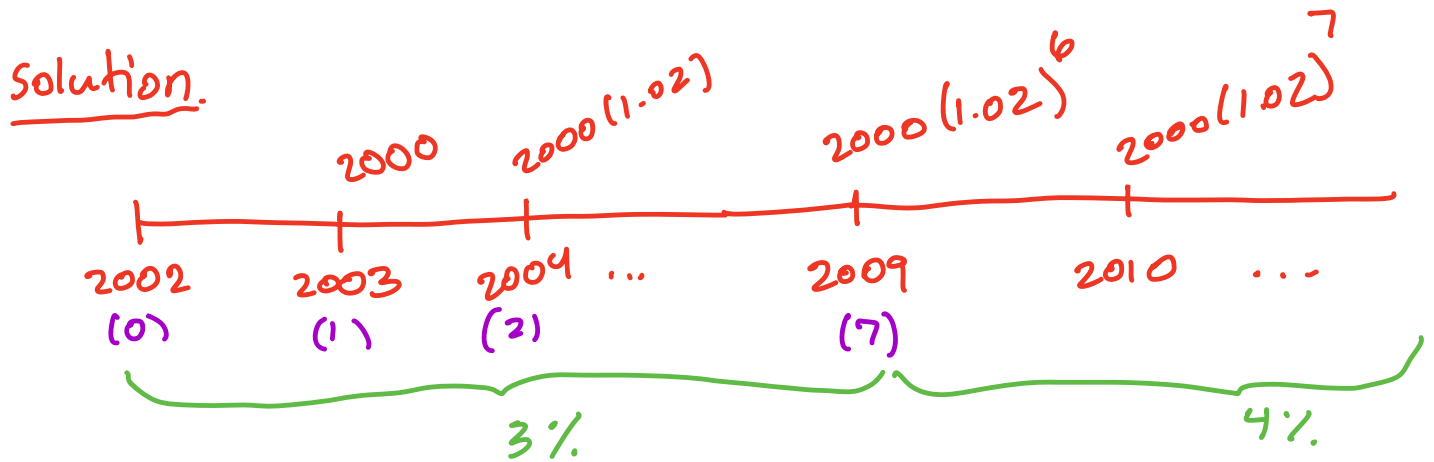
$$\text{Value at } t=0 \rightarrow 800 \left(\frac{1 - \left(\frac{1.03}{1.07} \right)^{25}}{1.07 - 1.03} \right)$$

$$= 12,284.46$$

Notice if $g=0$ we get back

$$P \left(\frac{1 - \left(\frac{1}{1+i} \right)^n}{i} \right) = a_{\overline{n}|i}$$

Problem: On January 1, 2002, Andrea inherits a perpetuity-immediate with annual payments. The first payment is \$2,000 and after that the payments increase by 2% each year. Find the value of this perpetuity on January 1, 2002 if the annual effective rate of interest is 3% from January 1, 2002 through January 1, 2009 and 4% thereafter.



$$a_{\overline{7}|.03} \text{ w/ growth } = .02, P = 2000$$

$$+$$

$$a_{\overline{\infty}|.04} \text{ with growth } = .02, P = 2000(1.02)^8$$

$$\rightarrow \text{times } (1.03)^{-8}$$

$$= 2000 \left(\frac{1 - \left(\frac{1.02}{1.03} \right)^7}{.01} \right) + 2000(1.04)^{-1}(1.02)^7(1.03)^{-7} \left(\frac{1}{1 - \frac{1.02}{1.04}} \right)$$

$$= 13,202.68688685 + 93,398.65656$$

$$= 106,601.34$$

3.9 Annuities with Arithmetic Progression payments

Consider an annuity lasting n periods with payments

$$j^{\text{th}} \text{ pmt} = P + Q(j-1)$$

$$j=1 \rightarrow P$$

$$j=2 \rightarrow P + Q$$

$$j=3 \rightarrow P + 2Q$$

$\vdots$

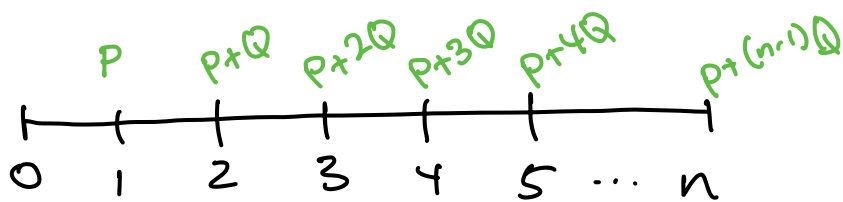
$$j=n \rightarrow P + (n-1)Q$$

Q is the additive change in pmts

We denote the present and future value by

$$(I_{P,Q,a})_{\overline{n}|i} \quad \text{and} \quad (I_{P,Q,a})_{\overline{n}|i}$$

Can we derive a formula for $(I_{P,Q,a})_{\overline{n}|i}$?



$$\begin{aligned}
 (I_{P,QS})_{\overline{n}i} &= P(1+i)^{n-1} \\
 &+ (P+Q)(1+i)^{n-2} \quad \begin{array}{l} \nearrow P(1+i)^{n-2} \\ \searrow Q(1+i)^{n-2} \end{array} \\
 &+ (P+2Q)(1+i)^{n-3} \quad \begin{array}{l} \nearrow P(1+i)^{n-3} \\ \searrow 2Q(1+i)^{n-3} \end{array} \\
 &+ \vdots \\
 &+ (P+(n-2)Q)(1+i)^1 \quad \begin{array}{l} \nearrow P(1+i) \\ \searrow (n-2)Q(1+i) \end{array} \\
 &+ (P+(n-1)Q)
 \end{aligned}$$

$$\begin{aligned}
 (1+i)(I_{P,QS})_{\overline{n}i} &= P(1+i)^n \\
 &+ (P+Q)(1+i)^{n-1} \quad \begin{array}{l} \nearrow P(1+i)^{n-1} \\ \searrow Q(1+i)^{n-1} \end{array} \\
 &+ (P+2Q)(1+i)^{n-2} \quad \begin{array}{l} \nearrow P(1+i)^{n-2} \\ \searrow 2Q(1+i)^{n-2} \end{array} \\
 &+ \vdots \\
 &+ (P+(n-2)Q)(1+i)^2 \quad \begin{array}{l} \nearrow P(1+i)^2 \\ \searrow (n-2)Q(1+i)^2 \end{array} \\
 &+ (P+(n-1)Q)(1+i) \quad \begin{array}{l} \nearrow P(1+i) \\ \searrow (n-1)Q(1+i) \end{array}
 \end{aligned}$$

lets subtract $(1+i)(I_{P,QS})_{ni} - (I_{P,QS})_{ni}$

$$i(I_{P,QS})_{ni} = P(1+i)^n$$

$$+ Q(1+i)^{n-1} - 0 = Q(1+i)^{n-1}$$

$$+ 2Q(1+i)^{n-2} - Q(1+i)^{n-2} = Q(1+i)^{n-2}$$

$$+ 3Q(1+i)^{n-3} - 2Q(1+i)^{n-3} = Q(1+i)^{n-3}$$

$\vdots$

$$+ (n-1)Q(1+i) - (n-2)Q(1+i) = Q(1+i)$$

$$- (P + (n-1)Q) = -P - nQ + Q$$

so we get $i(I_{P,QS})_{ni} =$

$$P[(1+i)^n - 1] + QS_{ni} - nQ$$

divide by i

$$(I_{P,Q} s)_{\overline{n}i} = P s_{\overline{n}i} + \frac{Q}{i} (s_{\overline{n}i} - n)$$

Multiplying by v^n to get present value:

$$(I_{P,Q} a)_{\overline{n}i} = P a_{\overline{n}i} + \frac{Q}{i} (a_{\overline{n}i} - nv^n)$$

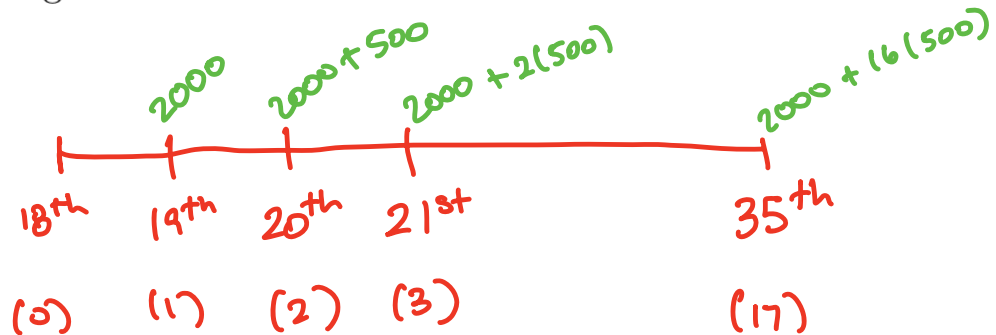
If $P=Q=1$ we write $(Ia)_{\overline{n}i}$

If $P=n$ and $Q=-1 \rightarrow (Da)_{\overline{n}i}$

pmts of $n, n-1, n-2, \dots, 2, 1$

Problem: Susan receives an annuity for her eighteenth birthday. The annuity pays \$2,000 on her nineteenth birthday and then has an annual payment on each of her birthdays through her thirty-fifth birthday. Each year the payments increase by \$500. Find the value of the annuity on Susan's eighteenth birthday, assuming the annual effective rate of interest is 4.2%.

Solution.



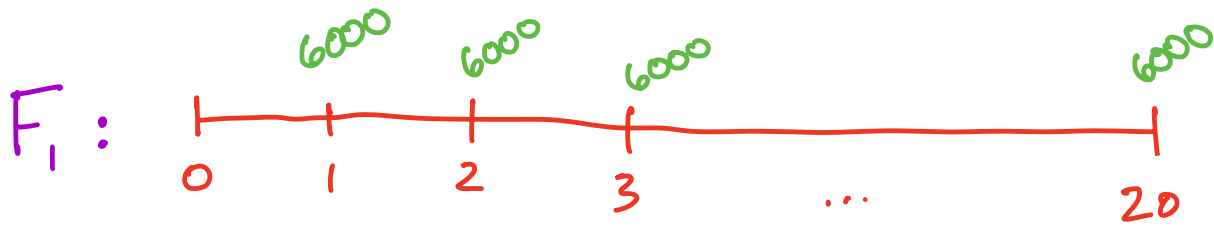
$$i = 4.2\% \quad (I_{2000, 500} a)_{\overline{17}|.042}$$

$$= 2000 a_{\overline{17}|.042} + \frac{500}{.042} (a_{\overline{17}|.042} - 17(1.042)^{-17})$$

$$= 66008.4287996$$

Problem: At the end of each year Mr. Dunn deposits \$6,000 into an investment fund. The fund pays out interest each year at an annual effective interest rate of 6%. Mr. Dunn is only able to reinvest this interest at an annual effective interest rate of 4%. Assuming Mr. Dunn continues to make his \$6,000 deposits for twenty years, that he always immediately deposits all the interest paid out into the 4% account, and that he makes no withdrawals, what is the accumulated value of Mr. Dunn's investments at the time of the last deposit?

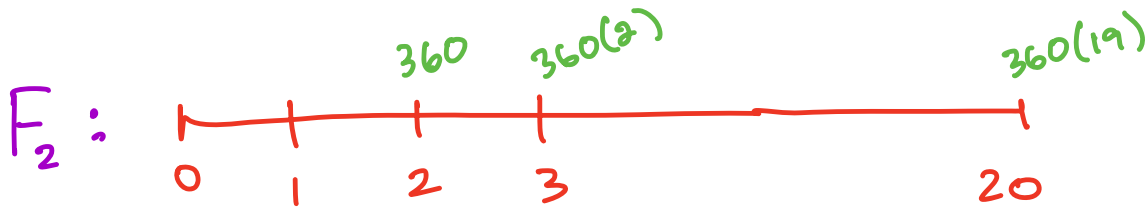
Solution.



Consider year



How much interest is earned? $K 6000 (.06)$
 $= 360K$

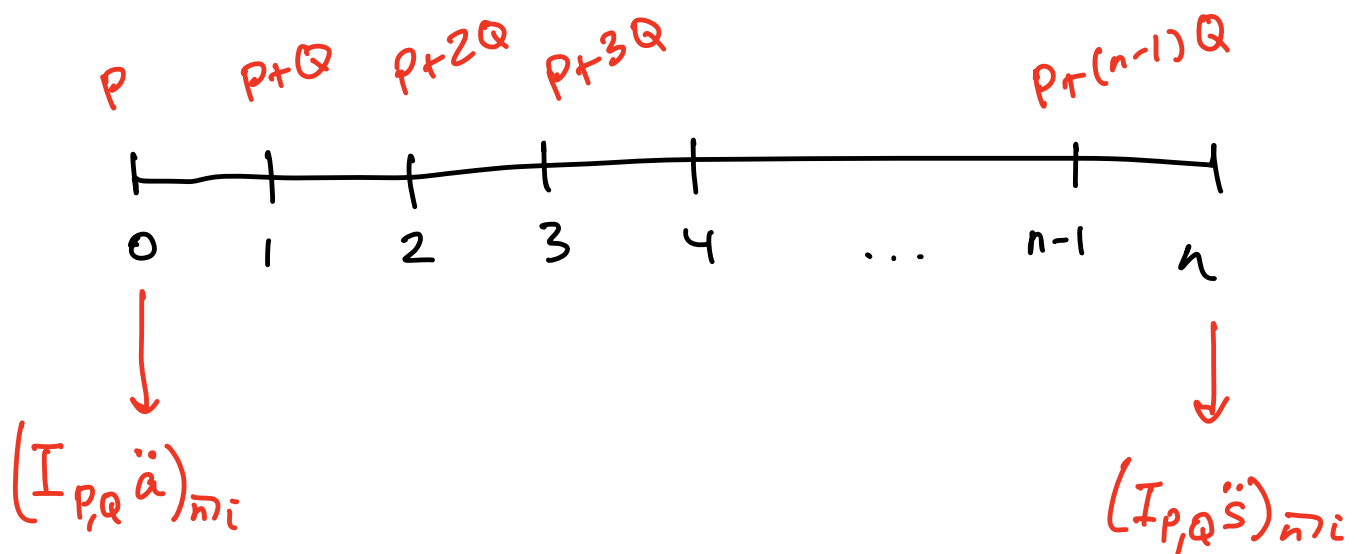
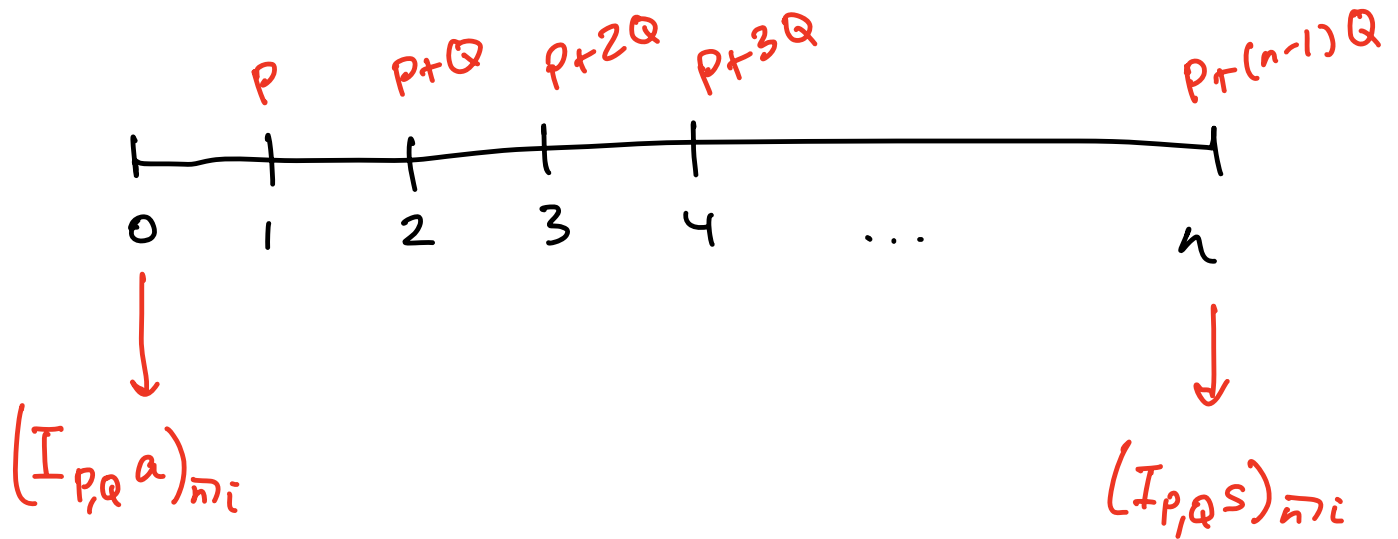


at time 20 $F_1 + F_2 = 120,000 + \left(I_{360,360} S \right)_{197.04}$

↖ No interest!

$$= 120,000 + 88,002.71 = 208,002.71$$

We can also have an arithmetic progression annuity, due!

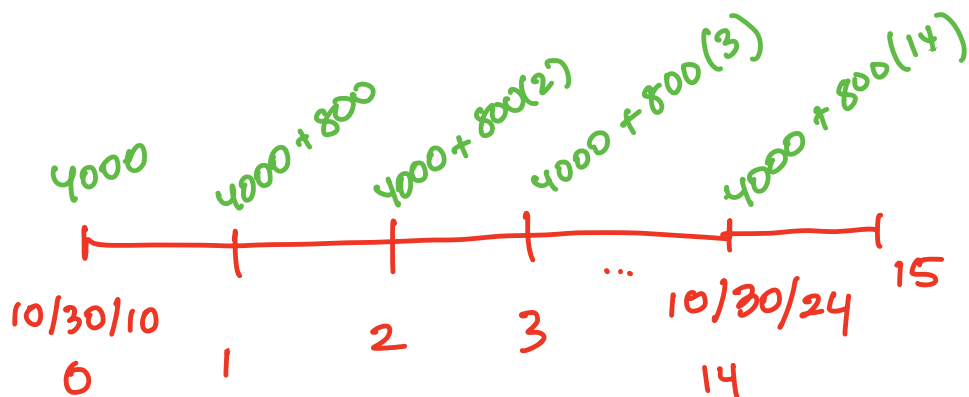


$$\text{Fact: } (I_{P,Q} \ddot{s})_{\overline{n}|i} = (1+i) (I_{P,Q} s)_{\overline{n}|i}$$

$$(I_{P,Q} \ddot{a})_{\overline{n}|i} = (1+i) (I_{P,Q} a)_{\overline{n}|i}$$

Problem: Alfonso has an annuity that will pay \$4,000 on October 30, 2010 and has annual payments through October 30, 2024. The payments increase by \$800 each year. As soon as Alfonso receives a payment, he will deposit it in a savings account with a 4% annual effective interest rate, and he will not make any withdrawals. What will be the balance in Alfonso's 4% account on October 30, 2025?

solution.



$$= \left(I_{4000, 800} \ddot{S} \right)_{\overline{15}|.04} = (1.04) \left[4000 s_{\overline{15}|.04} + \frac{800}{.04} \left(s_{\overline{15}|.04} - 15 \right) \right]$$

$$= 187,788.75$$

we can also have perpetuities with pmts in an arithmetic progression

Let's use limits to calculate

$$(I_{P, Q} a)_{\infty | i}$$

$$(I_{P,Q} a)_{\infty i} = \lim_{n \rightarrow \infty} P a_{ni} + \frac{Q}{i} (a_{ni} - n v^n)$$

$$= P a_{\infty i} + \frac{Q}{i} a_{\infty i} - \frac{Q}{i} (?)$$

$$\lim_{n \rightarrow \infty} \frac{n}{(1+i)^n} = \frac{\infty}{\infty} \xrightarrow{\text{L'ho pte's}} \frac{1}{(1+i)^n \cdot \ln(1+i)} = 0$$

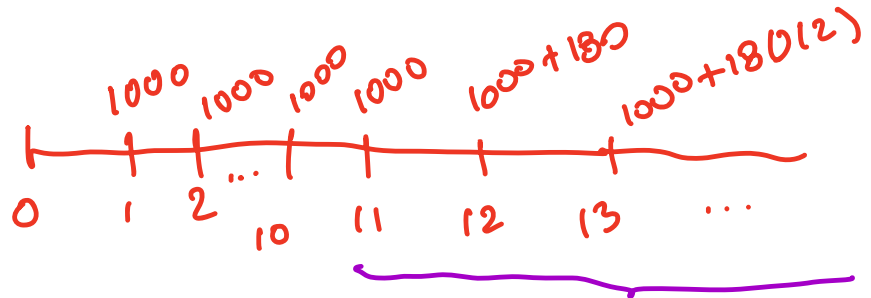
$$= \frac{P}{i} + \frac{Q}{i^2}$$

We also get $(I_{P,Q} \ddot{a})_{\infty i}$

$$= \frac{P}{d} + \frac{Q}{id}$$

Problem: Rafael purchases a perpetuity-immediate. The perpetuity pays \$1,000 at the end of each of the first eleven years and then has payments that increase by \$180 each year. So, the payment at time 11 is \$1,000, and the payment at time 12 is \$1,180. The purchase price was determined using a 5% annual effective interest rate. Find Rafael's purchase price.

Solution.



$i = 5\%$.

Find value at $t = 0$

$$(I_{1000, 180} a)_{\overline{\infty}|.05}$$

$$1000 a_{\overline{10}|.05} + (1.05)^{-10} (I_{1000, 180} a)_{\overline{\infty}|.05}$$

$$= 1000 \left(\frac{1 - \left(1 - \frac{.05}{1.05}\right)^{10}}{.05} \right) + (1.05)^{-10} \left(\frac{1000}{.05} + \frac{180}{.05^2} \right)$$

$$= 64,201.75$$